higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

WORKSHOP PRACTICE
NQF LEVEL 2

(12041012)

20 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 5 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Read ALL the questions carefully.
2. Answer ALL the questions.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

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QUESTION 1

1.1 Give TWO examples of unsafe conditions which include any change from accepted safety standards and that may cause accidents or incidents. (2)

1.2 List SIX environmental regulations that are compulsory in the workplace and which affect the safety and health of workers. (6)

1.3 An employer shall take the prescribed steps to ensure that a health and safety committee complies with the provisions of section 19 (4) and performs the duties assigned to it by subsection (1) and (2).

Fill in the missing words in the statement below and write only the answer next to the numbers (1.3.1–1.3.4) in your ANSWER BOOK.

A health and safety committee shall hold meetings as (1.3.1) ... as may be necessary, but at least once every (1.3.2) ... months at a (1.3.3) ... and (1.3.4) ... determined by the committee. (4 × 1) (4)

14 List the FOUR steps used by NOSA to obtain a star rating allocation for a business. (4)

15 What is the minimum number of employees a business shall have in his/her employment at any workplace before a health and safety representative has to be appointed as stipulated by the OHSA of 1993. (1)

16 Name THREE items of PPE that must be worn during an arc welding operation to prevent the operator from being injured by the welding sparks or ultra-violet rays. (3)

[20]

QUESTION 2

2.1 The immediate treatment of any casualty is very important. It is therefore necessary to monitor the casualty's vital signs.

Name TWO signs of life an aid worker must look out for. (2)

2.2 One of the most common causes of respiratory disturbance is choking caused by a foreign object lodged in a casualty's windpipe which he/she is unable to expel.

Keeping age in mind name the THREE possible treatments when someone is choking. (3)

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2.3 Name FOUR accredited occupational health or first-aid providers who may issue a valid certificate of competency to a first-aider in the workplace. (4)

2.4 List the THREE different ways by which a victim can be poisoned and give an example of each. (3 × 2) (6)

[15]

QUESTION 3

3.1 At what angle must a centre punch's point be? (1)

3.2 What makes a pin punch to malfunction? (2)

3.3 Identify the function of the following types of punches:

A Centre punch
B Pin punch
C Hollow punch (6)

3.4 Various hammers are used for different applications.

3.4.1 Name TWO types of hammers available. (2)

3.4.2 List FOUR procedures when working with a hammer. (4)

[15]

QUESTION 4

4.1 Name FOUR portable power tools an engineering technician will work with that can be AC or DC operated. (4)

4.2 List THREE safety elements to consider when maintaining and caring for the following power tools and accessories:

4.2.1 Grinding wheels (3)

4.2.2 Drill bits (3)

4.3 A jig saw is one of the power tools that is most commonly used in everyday life.

4.3.1 What is the function of a jigsaw? (2)

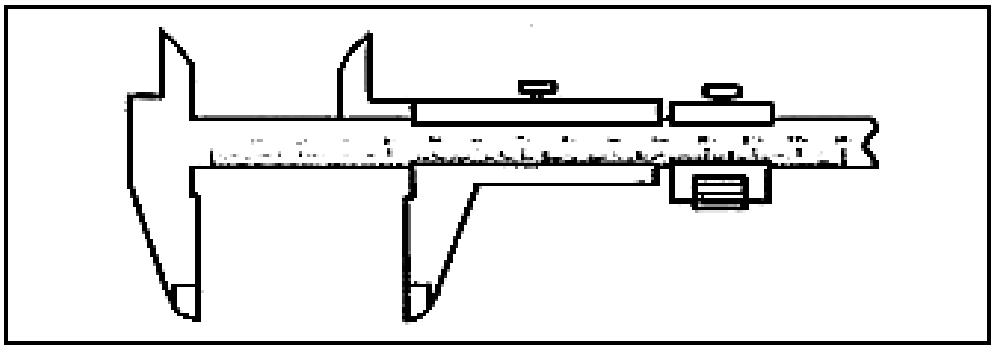
4.2.1 Which part of a jigsaw should never be touched when it is used? (1)

4.2.1 Which PPEs are recommended when using a jigsaw? (2)

[15]

QUESTION 5

5.1 5.1.1 Identify the engineering measuring equipment shown in Figure 1 below.



(1)

FIGURE 1

5.1.2 List FOUR checks that should be carried out on the measuring equipment in QUESTION 5.1.1 before using it. (4)

5.2 Name THREE safety precautions when using measuring tape. (3)

5.3 Name FOUR pocket-sized measuring tapes that are usually available in four main lengths. (4)

5.4 Explain how you would store a measuring tape. (3)

[15]

QUESTION 6

6.1 Explain the purpose of symbolic signs in the workplace. (3)

6.2 Name FIVE causes of fires. (5)

6.3 Name the type of chain block which consists of two pulleys with different diameters that are joined together. (1)

6.4 Name ONE other type of chain block you are familiar with. (1)

[10]

QUESTION 7

7.1 List FIVE features you should consider when choosing a soldering iron. (5)

7.2 List the steps that should be followed when treating a person who has received a burn from a hot soldering iron or a hot workpiece. (5)

[10]

TOTAL: 100

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